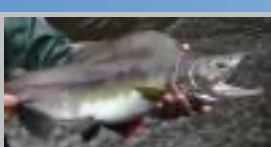


NAS - Nonindigenous Aquatic Species



***Oncorhynchus gorbuscha***

(Pink Salmon)

FishesNative Transplant

Collection Info

Point Map

Species Profile

Animated Map

← Image 1 of 3 →



Earl Steele 

*Oncorhynchus gorbuscha* (Walbaum, 1792)

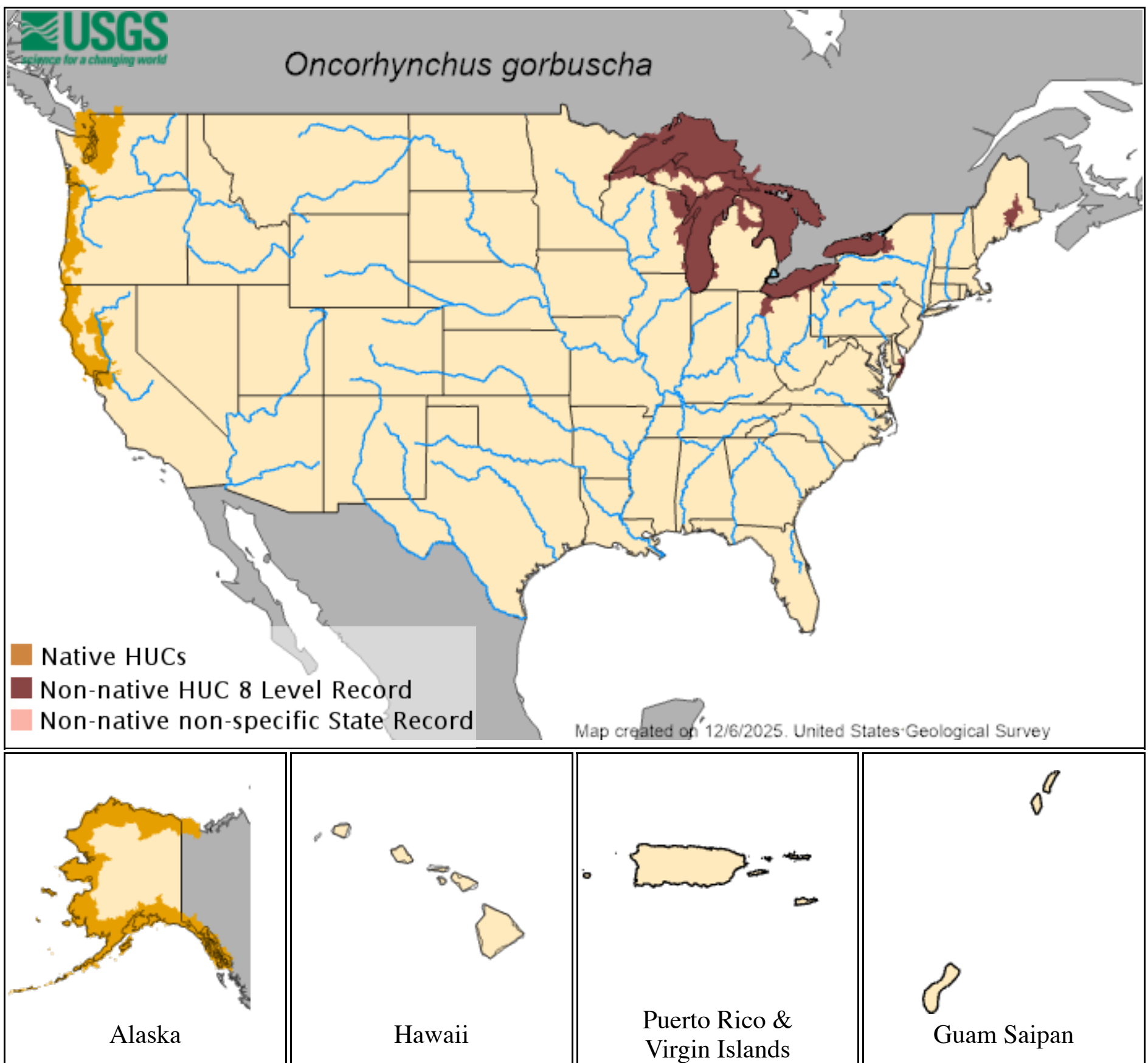
**Common name:** Pink Salmon

**Taxonomy:** available through 

**Identification:** *Oncorhynchus gorbuscha* has two dorsal fins including one adipose fin with 13–17 rays in its anal fins. It has a dark mouth and gums and is mostly silver overall, with large oval black spots on the tail and back (Michigan DNR 2003). During spawning, its sides and back are pinkish, giving this species its common name (Halpern et al. 2002). See also: Moyle (1976a); Scott and Crossman (1973); Wydoski and Whitney (1979); Morrow (1980); Eschmeyer et al. (1983); Page and Burr (1991).

**Size:** 76 cm

**Native Range:** Arctic and Pacific drainages from Mackenzie River delta, Northwest Territories, to Sacramento River drainage, California; occasionally as far south as La Jolla, southern California. Also in northeastern Asia (Page and Burr 1991).



Native range data for this species provided in part by [NatureServe](#)  
[Hydrologic Unit Codes \(HUCs\) Explained](#)  
Interactive maps: [Point Distribution Maps](#)

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of [HUCs](#) with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Oncorhynchus gorbuscha* are found [here](#).

| State | First Observed       | Last Observed        | Total HUCs with observations† | HUCs with observations†                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------|----------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IL    | <a href="#">1973</a> | <a href="#">1973</a> | 1                             | <a href="#">Lake Michigan</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| IN    | <a href="#">1973</a> | <a href="#">1973</a> | 1                             | <a href="#">Lake Michigan</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ME    | <a href="#">1914</a> | <a href="#">1973</a> | 2                             | <a href="#">New England Region</a> ; <a href="#">Penobscot River</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| MD    | <a href="#">1973</a> | <a href="#">1973</a> | 1                             | <a href="#">Chincoteague</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| MI    | <a href="#">1959</a> | <a href="#">2019</a> | 20                            | <a href="#">Au Sable</a> ; <a href="#">Betsie-Platte</a> ; <a href="#">Betsy-Chocolay</a> ; <a href="#">Black-Macatawa</a> ; <a href="#">Black-Presque Isle</a> ; <a href="#">Carp-Pine</a> ; <a href="#">Cedar-Ford</a> ; <a href="#">Cheboygan</a> ; <a href="#">Dead-Kelsey</a> ; <a href="#">Fishdam-Sturgeon</a> ; <a href="#">Keweenaw Peninsula</a> ; <a href="#">Lake Huron</a> ; <a href="#">Lake Michigan</a> ; <a href="#">Lake Superior</a> ; <a href="#">Lone Lake-Ocqueoc</a> ; <a href="#">Millecoquins Lake-Brevoort River</a> ; <a href="#">St. Marys</a> ; <a href="#">Sturgeon</a> ; <a href="#">Tahquamenon</a> ; <a href="#">Waiska</a> |
| MN    | <a href="#">1956</a> | <a href="#">2022</a> | 3                             | <a href="#">Baptism-Brule</a> ; <a href="#">Beaver-Lester</a> ; <a href="#">Lake Superior</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| NY    | <a href="#">1979</a> | <a href="#">1986</a> | 3                             | <a href="#">Lake Erie</a> ; <a href="#">Lake Ontario</a> ; <a href="#">Salmon-Sandy</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| OH    | <a href="#">1979</a> | <a href="#">1986</a> | 3                             | <a href="#">Ashtabula-Chagrin</a> ; <a href="#">Lake Erie</a> ; <a href="#">Sandusky</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| PA    | <a href="#">1979</a> | <a href="#">1979</a> | 2                             | <a href="#">Chautauqua-Conneaut</a> ; <a href="#">Lake Erie</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| WI    | <a href="#">1959</a> | <a href="#">2014</a> | 10                            | <a href="#">Beartrap-Nemadji</a> ; <a href="#">Door-Kewaunee</a> ; <a href="#">Lake Michigan</a> ; <a href="#">Lake Superior</a> ; <a href="#">Lower Fox</a> ; <a href="#">Manitowoc-Sheboygan</a> ; <a href="#">Menominee</a> ; <a href="#">Oconto</a> ; <a href="#">Peshtigo</a> ; <a href="#">Pike-Root</a>                                                                                                                                                                                                                                                                                                                                               |

Table last updated 12/8/2025

† Populations may not be currently present.

**Ecology:** Pink salmon are coldwater fish with a preferred temperature range of 5.6 to 14.6 °C, an optimal temperature of 10.1 °C, and an upper incipient lethal temperature of 25.8 °C. Once thought to require salt water to complete their life cycle, a population has become established in the Great Lakes which exhibits genetic variation from their anadromous counterparts that allows them to tolerate freshwater throughout their lives (Gharrett and Thomason, 2011). Lake Superior pink salmon are phenotypically similar to their Pacific Coast counterparts, the major difference being age structure and smaller size of adults. Approximately 90% of the adults mature as 2-year-olds while the remaining 10% mature as 3-year-olds. Pink salmon maturing as 3-year-olds have different growth patterns than those maturing at 2 years of age. Two-year-old pink salmon average 390 mm in length, which is about 30% smaller than 2-year-olds from the Pacific Ocean. Three-year-old females have a lower fecundity and a poorer egg quality than 2-year-old females (Bagdovitz et al. 1986).

In the Great Lakes, pink salmon move from open lake into rivers to spawn in late summer or early autumn and not all return to their natal river (Kwain and Rose 1986). With a life span of only two or sometimes three years, pink salmon typically spawn biennially (Kocik et al. 1991). After spawning, the female guards the nest and dies within a few days or weeks. Spawning males defend territories, and adults die soon after spawning. Eggs hatch from late December to late February depending on water temperatures. When the young mature enough to leave their gravel nest in late April or early May, they journey downstream in large schools. The young salmon reach adulthood in about 18 months with a size of two to seven pounds and 17 to 19 inches in length (Michigan DNR 2003).

**Means of Introduction:** This species was introduced accidentally into the Canadian side of Lake Superior in Thunder Bay in 1956, when about 100 salmon escaped while being transferred from a hatchery to a seaplane (Schumacher and Eddy 1960; Eddy and Underhill 1974). Another reference states that about 21,000 surplus fingerlings from the hatchery at Port Arthur, Ontario, were disposed of into the Current River, a tributary of Lake Superior (Becker 1983). The hatchery was raising them to stock in Hudson Bay (Eddy and Underhill 1974). The first evidence of reproduction was noted in 1959, when two second-generation fish were caught in Lake Superior, Minnesota (Schumacher and Eddy 1960; Schumacher and Hale 1962; Eddy and Underhill 1974). The species was first observed in a northern tributary of Lake Huron in 1969 (Parsons 1973). By 1979, it had spread throughout the Great Lakes (Cooper 1983). Emery (1981) gave an account of their spread through the Great Lakes and included a map showing the dispersal. The species was also intentionally stocked on the coasts of Maine and Maryland (Morrow 1980). Also see Schumacher and Eddy (1960), Schumacher and Hale (1962), Parsons (1973), Wagner (1976), Emery (1981), and Kwain and Lawrie (1981).

**Status:** Established population present throughout the Great Lakes (Emery 1981; Trautman 1981; Phillips et al. 1982; Cooper 1983). Populations in Maine and Maryland survived until about 1973, then disappeared (Morrow 1980).

**Impact of Introduction:** Pink salmon may displace native chubs by way of food competition. The introduced salmon also may compete with lake herring (Becker 1983). Individuals over one year old feed heavily on rainbow smelt and alewife, which are important components of the diets of other Great Lakes salmonids (Kocik et al. 1991).

**References** (click for full references)

Becker, G.C. 1983. Fishes of Wisconsin. University of Wisconsin Press, Madison, WI. 1052 pp. Available: <http://digital.library.wisc.edu/1711.dl/EcoNatRes.FishesWI>

Diana, J.S. 1990. Food habits of angler-caught salmonines in western Lake Huron. Journal of Great Lakes Research 16(2):271-278.

Halpern, T., N. Paulson, and J.T. Hatch. 2002. Pink salmon. Minnesota Department of Natural Resources. [http://www.gen.umn.edu/research/fish/fishes/pink\\_salmon.html](http://www.gen.umn.edu/research/fish/fishes/pink_salmon.html)

Kirkpatrick, N.S., D.W. Everitt, and B.I. Evans. 2007. Asymmetric hybridization of pink (*Oncorhynchus gorbuscha*) and Chinook (*O. tshawytscha*) salmon in the St. Marys River, Michigan. Journal of Great Lakes Research 33(2):358-365.

Kocik, J.F., and M.L. Jones. 1999. Pacific salmonines in the Great Lakes basin. In Taylor, W.W. and C.P. Ferreri, (Eds.). Great Lakes Fisheries Policy and Management: A Binational Perspective. Michigan State University Press, East Lansing, MI. pp. 455-488.

Kocik, J.F., and W.W. Taylor. 1987. Diet and movements of age-1+ pink salmon in western Lake Huron. Transactions of the American Fisheries Society 116(4):628-633.

Kocik, J.F., W.W. Taylor, and W.C. Wagner. 1991. Abundance, size, and recruitment of pink salmon (*Oncorhynchus gorbuscha*) in selected Michigan tributaries of the upper Great Lakes, 1984-1988. Journal of Great Lakes Research 17(2):203-213.

Kwain, W., and G.A. Rose. 1986. Spawning migration of Great Lakes pink salmon (*Oncorhynchus gorbuscha*): size and sex distributions, river entrance and exit. Journal of Great Lakes Research 12(2):101-108.

Michigan Department of Natural Resources (MIDNR). 2003. Pink salmon (*Oncorhynchus gorbuscha*). [http://www.michigan.gov/dnr/0,1607,7-153-10364\\_18958-45686--,00.html](http://www.michigan.gov/dnr/0,1607,7-153-10364_18958-45686--,00.html)

Tilmant, J.T. 1999. Management of nonindigenous aquatic fish in the U.S. National Park System. National Park Service. 50 pp.

**Other Resources:**  
[Pink Salmon \*Oncorhynchus gorbuscha\*](#) - State of Michigan

[Great Lakes Waterlife](#)

[Fishes of Wisconsin](#) - Wisconsin Sea Grant

[FishBase Summary](#)

**Author:** Fuller, P., J. Liebig, J. Larson, and A. Fusaro

**Revision Date:** 12/20/2019

**Peer Review Date:** 6/26/2014

**Citation Information:**  
Fuller, P., J. Liebig, J. Larson, and A. Fusaro, 2025, *Oncorhynchus gorbuscha* (Walbaum, 1792): U.S. Geological Survey, Nonindigenous Aquatic Species Database, Gainesville, FL., <https://nas.er.usgs.gov/queries/FactSheet.aspx?SpeciesID=906>, Revision Date: 12/20/2019, Peer Review Date: 6/26/2014, Access Date: 12/8/2025

This information is preliminary or provisional and is subject to revision. It is being provided to meet the need for timely best science. The information has not received final approval by the U.S. Geological Survey (USGS) and is provided on the condition that neither the USGS nor the U.S. Government shall be held liable for any damages resulting from the authorized or unauthorized use of the information.